

What are Capabilities ? How Can RL Agents Use Their Own Capabilities while learning?

Abstract

Consider a simple task: picking up a 10 kg box. How do you evaluate whether this task falls within your capability? Beyond the complex neural and perceptual processing involved, there are a few core evaluation steps. First, you understand what the task means (e.g., what it means to “pick up” an object). Using an internal world model, you imagine what the goal state s' would look like after performing the task. Then, you determine whether the transition from your current state s to the goal state s' is feasible, based on past experience and learned capabilities. This proposal aims to identify key, computationally implementable aspects of human capability estimation and translate them into mechanisms for reinforcement learning agents. This is relevant because standard RL exploration often wastes samples on actions that are physically impossible or ineffective, which becomes especially costly in real world settings (e.g., robotics). The proposal talks about various problems that occur due to not being able to learn capabilities.

1 Introduction

Reinforcement learning agents typically employ epsilon greedy exploration in algorithms like Q-Learning and SARSA, while policy gradient and actor critic methods (e.g., PPO, A2C, TRPO) rely on stochastic policies that sample actions from probability distributions Gaussians for continuous actions or categorical distributions for discrete ones providing inherent exploration without explicit ε greedy mechanisms. These approaches explore actions without any predetermination; the agent has no prior knowledge of what will happen when it takes action a . The only way to discover what action a can accomplish is to execute it in the environment. This concept has been effective for agent learning: agents receive high rewards from certain actions and prioritize them to maximize cumulative reward, while learning to avoid actions that yield low or zero reward.

But is this how we learn? If I ask you to pick up a small box, then a larger box, and then progressively heavier objects up to a 100 KG dumbbell, would you actually attempt to lift the 100 KG dumbbell? Of course not. This reveals a fundamental limitation that RL agents currently face a core obstacle on the path toward AGI. We need to understand how humans learn and explore actions. While we fully grasp something only after experiencing it, there exists a form of intelligent predetermination that occurs before or during exploration. Our goal is to formalize this predetermination intelligence that humans possess and transfer it to RL agents. The crucial observation here is that humans learn from success not merely about the specific states where actions were taken, nor just about immediate neighbors, but about their own underlying capabilities for the task. We need to expand what RL agents can learn from success. Similarly, consider learning from failure: if both an agent and a human attempt to lift 50KG, 45KG, and 40KG and fail at all three, a human can still accurately answer “can you lift 2 kg?” (yes), but can an RL agent? This falls under the category of learning from failures, which authors like Shiarlis et al. have addressed , they propose inverse reinforcement learning from failure (IRLF) which exploits both successful and failed demonstrations. The Proposal Identifies both problems: learning from failures (which has been discussed before) and learning

more effectively from successes (which to our knowledge has been addressed for the first time). Wang et al 2024 masks invalid actions which produce a transition to the same state. However, this approach has several critical limitations. First, an action may produce a transition to state s' but still might be infeasible. Second, it is not necessary that a transition via action a on state s that leads to the same state s (i.e., $P(s|s, a) = 1$) is always due to an infeasible action. For example, in car driving during a red light, the action “stay” leads to the same state but is clearly not infeasible.

We can develop a more sophisticated form of action masking that improves upon the current masking limitations. Key components of signals may include concepts like empowerment which Klyubin et al. 2005, Salge et al. 2014: Defined as $I(a; s' | s)$ which is mutual information between action and next state. Empowerment remains the core signal in our overall approach. We currently estimate 0 empowerment as a signal to infeasibility. Another concept in multi signal approach may also include affordance Khetarpal et al 2020: which refers to the set of action possibilities that an environment or object offers to an agent, given the agent’s capabilities and the current state of the environment. Affordance is one of the most relevant approaches but requires expert prior information or pretrained model.

A core component of human reasoning is the world model, which provides predictions of next states s' . Our RL agent will employ a partial world model inspired by Freeman et al 2019. The role of world model will be to provide estimated state transitions to pre determine the masking for future transitions.

We currently also expect that the agent would learn behaviour that reflect high level intelligence, for example when the agent learns some actions like picking are not feasible the agent will try other actions like push and achieve it’s overall goal of moving the object, in extreme cases when all actions of agent are infeasible it might try moving to some other state so that it gets better empowerment upon the overall task. An example would be that upon learning that objects beyond 1m are unreachable, the agent could learn to use a nearby stick to reach the object.

Moreover, imitation learning can be used to extract feasibility cues from human behavior (e.g., Hussein et al.). This aims to train an agent to imitate an expert or human demonstrator, and then analyze how the agent’s state action choices change when it tries to copy the demonstrator. This can provide concrete insight into how humans avoid infeasible actions with minimal exploration, and how these implicit capability judgements can be extracted into meaningful insights.

Key solutions in the proposal are:

- **Multi signal action masking:** develop a more sophisticated form of action masking using multiple signals, including empowerment and affordance style preconditions, to detect infeasible actions and saturated action components online.
- **Partial world model for lookahead:** learn a partial world model and use it to look n steps ahead, so the agent can predict infeasible transitions $s \rightarrow s'$ before executing actions, improving sample efficiency.
- **Imitation learning for infeasibility cues:** train an agent to imitate expert or human behavior and analyze which state action choices it makes, with the goal of extracting how humans avoid infeasible actions with minimal exploration.

2 Related Work

Dey et al 2023: Leverages predefined constraints to label states as safe or unsafe, then masks actions using membership probability. The assumption that constraints are available for every environment does not meet our goal of online learning. (Dalal et al. 2018, Achiam et al. 2017

CPO): Learn constraint boundaries from violations. Learning capability limits from observed boundary hits. Safe RL avoids unsafe states, while we identify actions that are physically impossible to execute. Abel et al. 2018 examine when two actions are equivalent in an MDP. Action equivalence $\approx$ both infeasible or same effect. Touati and Ollivier 2021 introduce the forward-backward (FB) representation for learning a single representation that enables optimal policies for any reward specified after training. Touati et al. 2023 systematically test zero-shot RL schemes including successor features and forward-backward representations. Their work on learning universal representations that transfer across tasks complements our objective of learning transferable capability models.